

Keep Your Cascade, Harden Its Boundaries

The evidence says the reader should keep the cascade they have built (speech-to-text → LLM → text-to-speech over WhatsApp voice notes), not replace it with a native speech-to-speech model, because the 2026 production consensus is exactly that architecture: **cascade with stage-overlapped streaming is the stated enterprise default, and native speech-to-speech is reserved for products where conversational naturalness is the product and the tool surface is shallow** — the opposite profile of a GDS travel advisor that must read fare rules, PNRs and availability from Amadeus ([LiveKit](#); [Gadium](#)). The components already in the stack (Deepgram Nova-3, ElevenLabs Scribe v2 and multilingual v2, Deepgram Aura-2) are all on the 2026 component menu that voice-agent platforms recommend, so the work is not vendor replacement but boundary hardening: alphanumeric record-locator capture runs at roughly **85% sequence accuracy** on the one vendor figure available, locale number normalization is described as the single most common production defect, and Spanish/French formality is something models demonstrably hedge on ([Speechmatics](#); [Gadium](#)). The most honest production benchmark for travel is PolyAI at Hopper, which **fully resolves 15% of call volume**, far below ElevenLabs' "up to 80%" deflection marketing ([PolyAI](#)). The Big Four publish no voice-specific guidance at all; what they do publish is a unit-economics warning — **KPMG finds 49% of organisations scaled back agent deployments once costs outran value and only 7% report established ROI** — and a governance prescription (ringfenced permissions, real escalation paths, token controls) that the reader's daily spend cap already partly implements ([ppc.land summary of KPMG Q2 2026](#)). Regulation already binds: **EU AI Act Article 50 disclosure applies from 2 August 2026 and was not postponed**, Meta requires an AI statement in the first WhatsApp message, and the EDPB treats voice as inherently biometric ([Goodwin](#); [EDPB Guidelines 02/2021](#)). The largest unknowns are whether the WhatsApp Business Calling API exposes a streaming media path at all, what any vendor's Spanish or French word error rate actually is, and who is liable when an automated advisor misquotes a fare — none of which the research could answer. Every figure in this report, without exception, comes from search-engine summaries of the cited pages rather than full reads, and must be re-verified before it carries weight in a decision.

How to read this evidence: nothing here was read at source

All five research threads hit the same wall. The environment's egress proxy blocked direct fetches of nearly every primary domain — vendor documentation ([livekit.com](#), [deepgram.com](#), [elevenlabs.io](#), [openai.com](#), [assemblyai.com](#)), academic hosts ([arxiv.org](#), [huggingface.co](#)), the Big Four ([deloitte.com](#), [pwc.com](#), [ey.com](#), [kpmg.com](#)), EU legal sources ([eur-lex.europa.eu](#), [cnil.fr](#), [artificialintelligenceact.eu](#)) and Meta's developer docs ([developers.facebook.com](#), [business.whatsapp.com](#)). Web search worked; web fetch did not. **Every number, date and quotation below therefore comes from a search-engine summary of the cited page, not from the page itself.** The URLs are recorded so the reader can re-verify, and the report preserves the researchers' own confidence tiers: items tagged *vendor claim* are marketing self-assessment; items tagged *secondary* come from aggregators, comparison blogs or trade press rather than the originating organisation; items tagged *[UNVERIFIED]* are load-bearing details the researchers explicitly could not confirm and which may be wrong. The researchers also excluded several arXiv identifiers surfaced by search that do not correspond to plausible preprints, on the assumption they were hallucinated index entries. Where two sources conflict on the same figure, both are given and the conflict is named rather than resolved.

The researchers sorted the evidence into three reliability bands that the reader should keep in mind throughout. The strongest tier is vendor-confirmed and specific (PolyAI's Hopper and Golden Nugget figures, OpenAI's gpt-realtime token prices, Azure Voice Live GA at Build 2026, AssemblyAI's 18-language code-switching claim, Parloa's simulation pipeline) or multiply corroborated official material (Article 50's 2 August 2026 date, Article 5 since 2 February 2025, the Commission's antitrust press releases against Meta, *SCHUFA*). The middle tier is vendor marketing with no measurement behind it (ElevenLabs' 80% deflection, Retell's "unlimited concurrency," Speechmatics' and AssemblyAI's mutually exclusive "#1" claims, Gadium's ELO advantages). The weakest tier is practitioner-blog numbers with no traceable methodology — the 27% → 5% hallucination reduction, the 300 ms/600 ms/1.5 s caller-behaviour thresholds, the 32% → 8% abandonment claim — plus a set of specific low-confidence legal details: the 2 December 2026 watermarking transition, Anthropic's "Covered Models" retention, the 512 KB WhatsApp play-icon threshold, and the CNIL "first seconds of a call" rule.

Production voice companies run a component menu, and cascade is the default

The 2026 market has converged on a small, named set of interchangeable parts rather than on a single stack. For speech-to-text the recurring production names are **Deepgram Nova-3**, **AssemblyAI Universal-3.5 Pro** (preferred where multilingual or diarization matters) and **Speechmatics**; for text-to-speech, **ElevenLabs Flash v2.5**, **Cartesia Sonic-3/3.5/3.6** and **Deepgram Aura-2**; for the LLM, **OpenAI GPT-4.1-mini** as the default voice-agent brain with **Claude Sonnet 4.5** where long context matters; and for native speech-to-speech, **OpenAI gpt-realtime-2.1**, **Azure Voice Live** (GA for Foundry Prompt Agents as of Build 2026) and **Google Gemini Live** ([AssemblyAI](#); [Microsoft](#); [OpenAI](#)). The researchers stress an important caveat on this list: **no vendor stated, on any page they could verify, which STT or TTS it runs inside its own hosted agent product**. ElevenLabs Agents' default ASR, Vapi's defaults, Retell's internal stack and Sierra's underlying models are all undisclosed; the component recommendations above are third-party guidance, not vendor disclosure. The one genuinely vendor-confirmed model disclosure found is Parloa, which uses **GPT-4.1 and GPT-5-mini to simulate customer interactions before launch and then scores them with an LLM-as-judge plus deterministic rules** ([OpenAI customer story](#), summary only). That Parloa pattern — a cheap fast model in the live path, a stronger model offline for synthetic-traffic testing — is directly transferable to a travel advisor.

On orchestration the sources are unusually aligned. LiveKit's stated recommendation is to "default to cascade (STT → LLM → TTS); reach for speech-to-speech only where naturalness is the product," with a **hybrid — S2S for chit-chat, cascade for tool-calling turns — described as the pattern most production agents converge on** ([LiveKit](#); echoed by [Softcery](#)). Gadium's 2026 architecture comparison says the cascade is "the enterprise default" for anything needing auditability, compliance, provider flexibility, deep tool use and predictable cost, because "text at every boundary gives you logs, swappable components, and moderation hooks," while S2S means "accepting opaque failures, fewer vendors, and token cost that grows with call length" ([Gadium](#)). Modulate makes the same case on auditability alone ([Modulate](#)). The open-source framework choices are **Pipecat** (Daily; sub-500 ms interruptible turns, mid-stream function calling, transcript inspection) and **LiveKit Agents**, with Bedrock and Vertex as the hyperscaler alternatives; notably, **no named production users of Pipecat were found**, only sector-level claims ([Pipecat at GitHub](#); [WebRTC.ventures](#)).

The travel deployments that actually publish numbers are few and they are sobering. **PolyAI at Hopper fully resolves 15% of call volume**, and Hopper's VP of Customer Experience explicitly targets "resolution over containment," rejecting containment as the KPI ([PolyAI](#)). **PolyAI at Golden Nugget handles 34% of central-reservations calls**, averaging 300+ completed reservations per week with card capture within two months of launch ([PolyAI](#)). The gap between the two is intent complexity: hotel reservations are a narrow, high-intent task, whereas airline and GDS servicing (irregular operations, fare rules, involuntary reissue) sits at Hopper's end of the difficulty curve. Airline-side evidence is much weaker. Lufthansa has a Cognigy case study but the researchers could not recover its channel, languages, go-live date or any metric ([NiCE Cognigy](#), blocked); Iberia runs a dedicated WhatsApp assistant for booking management, baggage fees and flight status "supported by voice assistant integrations," which is the closest published analogue to the reader's product, but the voice phrase comes from low-quality trade press ([Travel And Tour World](#)). On current evidence **no airline anywhere has a verifiably published production voice agent**; airline activity is chat, self-service and agentic distribution. On the GDS side, Sabre doubled its agentic API/MCP partners from 30 to 60 and shipped end-to-end agentic booking with Mindtrip and PayPal in Q2 2026, while **Amadeus, Expedia and Booking.com have largely kept core systems closed or partner-limited** ([Skift](#)). One planning constraint is concrete: **Amadeus Altéa and Sabre integrations require partner agreements and take 6–12 weeks to provision** ([ValueStream AI](#), secondary).

Cost is where the marketing gap is widest. Advertised platform rates (Vapi \$0.05/min, Retell \$0.07–0.12/min) exclude the STT, LLM, TTS and telephony that get wired in, and **assembled Vapi production setups "commonly land around \$0.30 to \$0.33 per minute"** ([Medium/Automation Labs](#); corroborated in spirit by [Klariqo](#)). Raw component cost for STT+LLM+TTS is put at \$0.007–\$0.091 per conversation minute depending on model choice ([AssemblyAI](#)). Native S2S is priced per audio token — **gpt-realtime at \$32/M audio input and \$64/M audio output tokens, the mini at \$10/\$20** ([OpenAI](#); [eesel](#)) — and the resulting per-minute figures conflict sharply: one analysis puts all-in Realtime at **\$0.25–0.35/min** ([TokenMix](#)), while a 4,000-session measurement reports **\$0.06–0.11/min on gpt-realtime-2.1 once prompt caching works, and \$0.18–0.46/min for long calls without it** ([HackerNoon](#)). A third source puts a cascade at ~\$0.13/min against uncached realtime at ~\$0.32/min ([Auto Interview AI](#)). The structural reason is that S2S keeps the whole audio history in context on every turn, so cost scales with conversation length and caching is what makes it competitive — and caching is fragile ([Gradium](#)). For the reader's trilingual product, Deepgram bills multilingual streaming at a premium over monolingual; the two pricing summaries disagree on absolute levels (**\$0.0077 vs \$0.0092/min** per [ConvertAudioToText](#); **\$0.0048 vs \$0.0058/min** per [DIY AI](#)) but agree on the roughly 20% multilingual uplift, which is second-order because STT is a small share of total cost.

Real-time speech floors at 600 ms–1.2 s, and voice notes sit outside that budget

The three architectures are best understood as three different failure surfaces rather than three points on a latency line. A non-streaming cascade lands in seconds; a well-tuned streaming cascade lands around **600 ms–1.2 s voice-to-voice**; native S2S has a claimed floor of 300–500 ms time-to-first-audio but in independent measurement often lands at 0.7–3 s depending on how much reasoning it does. Two independent stage budgets agree closely: **"100–300 ms on speech-to-text, 350–1,000 ms on LLM inference, 90–200 ms on text-to-speech, and 50–200 ms on network**, totaling 600 ms to 1.7 seconds" ([Gradium](#)), and a Pipecat-oriented budget of STT 100–300 ms, LLM TTFT 200–600 ms, TTS first-audio 150–400 ms, network 20–100 ms ([Future AGI](#)). The dominant term in every budget is **LLM time-to-first-**

token, not the speech stages — which means the cheapest large latency win is a smaller or faster LLM or speculative first-sentence generation, not a TTS vendor swap. Vendors compete loudly on the 90 ms TTS number that is roughly 10% of the budget. On the S2S side, the Artificial Analysis leaderboard measured Gemini 3.1 Flash Live at **0.96 s TTFA at minimal reasoning effort rising to 2.99 s at high effort**, and Grok Voice at 0.70 s, which contradicts the clean "S2S floor = 300–500 ms" narrative; the floor figure describes the model's audio-generation floor, not the API round-trip a developer measures ([Artificial Analysis](#)).

The S2S field as of September 2026 has three serious hosted players and the newest one leads. **Google's Gemini 3.8 Live and 3.8 Live Extended Thinking (released 15 September 2026)** claim 97 supported languages with in-conversation switching, rank **#1 on the Artificial Analysis Speech-to-Speech Quality Index at 82.6**, score **68.6% on Sierra's τ -Voice** (up from 37.7% for the previous generation) but only **35.1% on τ -Voice-banking**, and add asynchronous function calling so tool calls no longer freeze the audio stream ([Google blog](#); [MarkTechPost](#)). Its hourly cost is reported as both **\$1.38/hour** ([Pasquale Pillitteri](#)) and **\$0.84/hour** ([MarkTechPost](#)) — the two figures conflict and Google's pricing page could not be reached. **OpenAI gpt-realtime-2.1** (shipped around July 2026) added better alphanumeric recognition, silence/noise handling and interruption behaviour, plus a distilled reasoning mini, but **supports function calling without structured outputs**, with guidance to "use a chained path when a strict structured intermediate contract is required" ([TechTimes](#); [benchr.org](#)). **Amazon Nova 2 Sonic** is the cheapest hosted option at roughly \$0.015/min with seven languages and async tool calling ([llm-stats](#), secondary). The τ -Voice-banking figure is the single most important S2S number for a travel advisor: even the best model fails two-thirds of voiced multi-step transactional tasks in a banking domain, and travel servicing is structurally similar. "Intelligence degradation" under audio input is a named, benchmarked phenomenon, and Big Bench Audio found that a Whisper → GPT-4o → TTS pipeline "shows minimal performance degradation compared to pure text processing" — the cascade preserves reasoning ([S2SBench](#); [Artificial Analysis/HuggingFace](#)), though Gemini 3.8 Live's 97.7% on that benchmark suggests the gap has largely closed at the frontier.

Streaming ASR has bifurcated into fast-transcript models and "conversational" models that fold end-of-turn detection into the recogniser. **Deepgram Flux** claims median end-of-turn detection under 300 ms (p95 1.5 s), saving 200–600 ms versus STT+VAD, with Coval cited as validating no accuracy trade-off ([Deepgram](#); [Deepgram citing Coval](#)); **AssemblyAI Universal-3.5 Pro Realtime** is reported at 6.99% pooled WER on the Pipecat agent benchmark versus 15.58% for Flux ([ChatGate](#), AssemblyAI-favourable). These two claims cannot both be the whole story; they measure different things on different data and neither should be treated as settled. Whisper is architecturally hostile to streaming — its encoder processes 30-second windows, so naive 1-second streaming re-runs the encoder every second and practical chunk-and-stitch latency is 5–10 s on consumer hardware ([arXiv 2506.12154](#); [SayToWords](#)). That disqualifies Whisper for a sub-second interactive agent but leaves it perfectly adequate for the reader's asynchronous voice-note cascade, where a multi-second transcription is invisible.

Turn-taking has moved off silence thresholds onto learned models: a Silero-class VAD gates audio, a small end-of-turn model decides whether the user has finished, and a classifier separates barge-in from backchannel. **Pipecat's Smart Turn v3 is ~8M parameters, int8, runs on the raw waveform in under 60 ms on cloud vCPUs, covers 23 languages and is fully open source** ([HuggingFace](#); [kwindla](#)); LiveKit's Turn Detector reads the last four turns of text and can classify "mm-hmm" versus true interruption ([Zylos Research](#)). A travel caller who thinks aloud ("we want to go to... uh... somewhere warm in, let's say, March") is precisely the case where naive silence-threshold endpointing interrupts, so semantic endpointing is not optional if the reader ever goes real-time. TTS latency is the most misleadingly marketed stage: vendor

time-to-first-byte claims cluster at 40–90 ms while independent P50 measurements of the same products cluster at 180–320 ms (**Cartesia Sonic-3 188 ms, ElevenLabs Flash v2.5 288 ms, Deepgram Aura-2 313 ms**), and the same benchmark flags **ElevenLabs Multilingual v2 and OpenAI tts-1-hd as showing "high variance unsuitable for real-time voice agents"** ([Gradium](#), run by a TTS vendor; Coval's framing is that vendor TTS benchmarks are not comparable, [Coval](#)). OpenAI publishes no latency specification for its standard TTS API at all ([Lushbinary](#)). PlayHT/PlayAI is reported as being wound down after Meta's acquisition and should be treated as deprecated ([SpeechGeneration AI](#)).

How much latency users tolerate is less settled than the industry implies. The human baseline is real and peer-reviewed: across ten languages the modal response offset is 0 ms and medians run 0 to +300 ms ([Stivers et al., PNAS 2009](#)). The operational thresholds that circulate (300 ms feels human, 600 ms reverts to touch-tone, 1.5 s hangs up) are practitioner heuristics with **no published study behind them** ([Forasoft](#); [Pa rloa](#)). The best controlled evidence found — a study of LLM-agent response delay — found users **preferred agents at 1.5 s latency (the low condition), with 46% naming fast response as the main reason for preferring an agent** ([arXiv 2507.22352](#)), which suggests the marginal return on grinding from 900 ms to 600 ms is smaller than the marketing implies and that consistency (tail latency) and not interrupting matter more than the median. No study of latency tolerance in asynchronous voice-note interaction exists — arguably the most important missing datum for the reader.

Transport is the hidden precondition of the whole migration question. WebRTC is the correct default for real-time voice (UDP, jitter-adaptive, Opus FEC), WebSockets on TCP are exposed to head-of-line blocking spikes of 100–300 ms, and SIP/PSTN adds RTT plus transcoding with real-world tail latency such as a reported **~325 ms one-way delay on a Twilio → LiveKit SIP trunk with both ends in US-Ashburn** ([LiveKit](#); [LiveKit Community](#)). For the reader specifically, **WhatsApp voice notes have no real-time transport at all**; the WhatsApp Business Calling API (launched 15 July 2025) is the only path to a streaming agent on that channel, access is still gated to businesses with established Meta partnerships or larger BSPs on early access ([Hyperleap](#)), eligibility reportedly requires a messaging limit of **at least 2,000 business-initiated conversations per rolling 24 hours** with call-permission requests capped at one per 24 hours and two per 7 days ([respond.io](#); [Wati](#)), and — critically — **whether the Calling API supports streaming audio for a machine agent, as opposed to human-operated calls, is unaddressed in every source found**. If the reader does get there, the transport budget resembles telephony rather than browser WebRTC, so sub-500 ms marketing targets are unreachable and an **800 ms–1.2 s P95 SLO is the realistic target**.

Spanish and French turn on codes, register and normalization, not headline WER

On clean read speech the top vendors are effectively tied. The English leaderboards are crowded within 1–2 points, and secondary comparisons put Nova-3, AssemblyAI Universal-3 Pro, gpt-4o-transcribe, ElevenLabs Scribe v2 and Microsoft MAI-Transcribe-1 **within 1–2 percentage points of each other on LibriSpeech and FLEURS** ([Coval](#); [VexaScribe](#), aggregator tier). The reader's ElevenLabs Scribe v2 Realtime is reported at **93.5% accuracy on FLEURS across 30 languages** ([futureagi](#), secondary), Whisper large-v3 at 4–6% WER for Spanish ([VexaScribe](#), aggregator), and Deepgram Nova-3 at a median 5.26% batch / 6.84% streaming on a general corpus ([futureagi](#)). Deepgram's own Spanish/French evidence is a **preference study against Whisper ("up to 8:1 preference ratios")** and relative WER gains over Nova-2, not a public-benchmark WER, which makes it weaker evidence than ElevenLabs' or AssemblyAI's FLEURS runs even if true ([Deepgram](#)); Spanish and French are also newer additions to Nova-3 than

English ([Deepgram](#)). The critical caveat is that FLEURS-style scores (4–6%) and agent-conversation scores (14–16%, e.g. Universal-2's 14.5% on a streaming comparison) differ by roughly 3x, so any comparison mixing the two is meaningless. **No verified per-language Spanish or French WER table for 2026 was obtainable** — the three sources that hold one (AssemblyAI's benchmarks page with FLEURS runs dated 25 August–3 September 2026, the Open ASR Leaderboard's multilingual track, and its paper) were all blocked ([AssemblyAI benchmarks](#); [Hugging Face](#); [arXiv 2510.06961](#)).

Code-switching is the requirement that actually narrows the field. Latin American Spanish and French speakers routinely mix English travel terminology (PNR, boarding pass, layover, e-ticket), and destination names are in the other language by construction. Only two vendors explicitly claim **single-pass native code-switching: AssemblyAI Universal-3.5 Pro across 18 languages "with no language-pair parameter and no separate routing step," and Speechmatics Melia across 55+ languages** ([Assembly AI](#); [Speechmatics](#), competitive page). Deepgram's multilingual mode is billed as a distinct tier, implying a different mechanism. Utterance-level language identification "fails completely on intra-sentential switches" and is highly likely to fail below about 2 seconds of audio ([Gladia](#); [USPTO 12057102](#)). The practitioner thresholds in circulation — require a 3-word minimum and confidence above 0.85 with a 500 ms debounce before switching language, retry with a backup model when STT confidence drops below 0.7, route to a human below 95% in high-stakes cases — are anecdotes from blog posts, usable as starting values but not as evidence ([AlterSquare](#); [Deepgram](#)). The worst LID case for a travel agent is the common one: a short opening turn like "Hola, Madrid, el martes" mixing a greeting, a proper noun and a date. Sub-2-second first turns should not be trusted for language routing; let the caller pick or confirm.

Alphanumerics are the failure mode that will actually hurt. **Speechmatics reports 96.9% sequence accuracy on character strings, 98.0% on digits, but only 85.4% on mixed alphanumerics** — the mixed case is exactly a record locator or fare basis, implying roughly one in seven spoken PNRs is wrong on first pass ([Speechmatics](#), vendor). AssemblyAI reports a **15.31% entity error rate for its own Universal-3.5 Pro Realtime on agent-conversation audio** and a 16.7% missed-entity rate for a competitor, and argues general WER is misleading for voice agents because it hides errors on confirmation codes, phone numbers and amounts ([AssemblyAI](#), vendor-run in both directions). Deepgram Nova-3 offers multilingual keyterm prompting as a biasing mechanism for exactly these tokens ([Deepgram](#)), and gpt-realtime-2.1's headline improvement was alphanumeric recognition, which confirms this is an industry-wide weak spot. The MT literature and the production write-ups point the same way on what to do about it: **extract candidate codes with format regexes (IATA [A-Z]{3}, six-character PNR, ISO dates, currency amounts), validate against live reference data or the PNR itself, replace them with placeholders before translation, re-substitute after, and re-verify before TTS** ([EmergentMind](#); [arXiv 2310.05824](#)). Verbatim glossary constraints are the wrong tool for anything inflected — "boarding pass" → *tarjeta de embarque* must agree in number and gender — so lemma-level guidance belongs on domain vocabulary and verbatim freezing only on opaque tokens ([RWS](#); [Translate With Care, ACL 2025](#)). Read-back confirmation must use each locale's own spelling alphabet, not the English NATO one.

Two further weaknesses are specific to these languages. First, **locale text normalization is described as the single most common defect in production voice agents**: "1.234" is *one point two three four* in English but *one thousand two hundred thirty-four* in Spanish, date order flips, and many engines normalize per language rather than per locale ([Gradium](#); [Inworld](#)). Second, formality. Manual inspection of LLM translations finds models "use *tu* almost exclusively for informal targets, while formal translations are more varied and models often avoid second-person forms entirely" — an avoidance strategy that reads as stilted, not correct formal register ([ScienceDirect](#)); inclusive French forms are rarely produced and "prompting

alone does not close the gap" ([arXiv 2506.15676](#)); and non-English-centric pairs consistently score lower on COMET than English-centric ones across 15 models and 22 pairs, which argues for pivoting ES ↔ FR through English if that direction is ever needed ([arXiv 2601.04036](#)). The tú/usted and tu/vous choice is therefore a product decision to pin per locale in the system prompt and verify in evaluation, not something the model will infer. Regional Spanish is well served commercially (Castilian, Mexican as the de facto neutral LatAm voice, Argentine, Colombian and others are advertised), but **fr-CA is markedly less covered than es-MX/es-ES in the 2026 write-ups**, and no measured MOS for any commercial Spanish or French voice was found ([SpeechGen](#), marketing). On the TTS side the choice is between one multilingual model that holds a single voice identity across languages (ElevenLabs Eleven v3 on one model ID for 74 languages; Inworld TTS-2 claims 200+ from one clone) and per-locale catalogues (Google Chirp 3 HD, Azure) that give explicit es-ES/es-MX/fr-FR/fr-CA control ([ElevenLabs](#); [Inworld](#), vendor ranking).

Real-time translation has become a single API call in 2026 if the reader wants it: **OpenAI gpt-realtime-translate (May 2026; 70+ input, 13 output languages) measured at a median 711 ms to first translated audio versus ~2.9 s for Google's gemini-3.5-live-translate-preview** on one single-source comparison ([OpenAI docs](#); [LiveLingo](#), methodology unverified), and **DeepL Voice-to-Voice (April 2026)** covering all 24 official EU languages with voice preservation in 14 ([DeepL](#)). The controllability argument cuts against them for this product: a single-call translator collapses ASR+MT+TTS and removes the intermediate text where placeholders, glossaries and PNR validation live. The quality comparison in circulation (cascaded Whisper+NLLB at 21.6 BLEU many-to-many versus 15.8 for SeamlessM4T v2, direct winning into English) rests on 2023/24 models and may not survive 2026 systems ([Deepgram](#)). Monitoring should be per language, because "regressions hide in aggregates": per-language WER targets, entity error rate as a first-class metric, COMET plus LLM-judge rubrics rather than BLEU for translation, native-speaker MOS panels for synthesis, and P50/P90/P99 latency by language ([Hamming](#); [futureagi](#)).

The Big Four publish governance and unit economics, not voice engineering

None of the four firms has a publicly identifiable flagship report on voice agents. Voice appears only as a use-case slice inside agentic-AI surveys under "customer service," and searches for Big Four guidance on build-vs-buy for voice, vendor lock-in, multi-model strategy, multilingual customer service or EU AI Act readiness returned only voice-vendor blogs — the researchers report this as a genuine absence, not a search failure. A buyer therefore cannot cite a Big Four framework to justify a voice architecture to a board; they can cite the firms on the governance wrapper around it. The commercial-interest caveat applies throughout: each firm publishes a large adoption number that creates urgency, a governance or maturity deficit that creates a gap, and a service line that closes it.

The numbers that matter are the sceptical ones, and they are recent. **KPMG's Global AI Quarterly Pulse for Q2 2026 (2,145 leaders, 20 countries) finds 49% of organisations scaled back, narrowed, delayed or paused AI agent deployments after expected costs began to outweigh value, and only 7% report established ROI; organisations with full visibility into AI operating costs report established ROI at 15% versus 3% for those without, yet only about a third have that visibility and only 36% have implemented direct token or usage controls** ([ppc.land](#); [Enterprise DNA](#); primary [KPMG](#), blocked). PwC's 2025 AI Agent Survey reads far more optimistically — **79% say agents are being adopted, 66% of adopters report measurable productivity value, 88% plan budget increases** — but the two are not contradictory: PwC asks about perceived productivity gain, KPMG about established, booked ROI, and the

market's mood shifted between the two fieldwork dates ([PwC](#)). KPMG's later, more sceptical data should carry more weight for a 2026 decision. EY's Responsible AI Pulse (975 C-suite leaders, 21 countries, August–September 2025) reports **99% of organisations suffered financial losses from AI-related risks, 64% exceeding US\$1m, while 87% have policies ringfencing what agents may do and 80% have incident escalation procedures** — high enough to suggest scoped permissions and escalation paths are procurement table stakes, not differentiators ([EY Global](#); [EY Singapore](#)). Deloitte's contribution is the diagnosis that "agentic AI is scaling faster than guardrails," with **only one in five companies holding a mature governance model for autonomous agents** — wording and denominator unverified against the primary page ([Deloitte Insights](#)). Mapped onto a stack, EY owns containment and escalation, Deloitte the maturity narrative, KPMG FinOps-for-agents, and PwC operating-model redesign; a travel voice deployment needs all four layers and no single firm covers them.

Deloitte is the only firm with substantive travel and aviation AI research, and it is a headwind for cost-to-serve positioning. In its **2026 Airline CEO Survey (21 CEOs, April–May 2026), AI overtook data analytics as the top technology priority, but over 80% of CEOs identified revenue management and dynamic pricing — not customer service — as the most significant AI impact area**, with talent shortages and legacy integration as the main barriers ([Deloitte Insights](#)). A voice agent sold to airlines on cost-to-serve competes for budget against revenue-generating AI and probably loses; the stronger positioning is revenue — upsell, ancillary attach, re-accommodation that preserves fare value. The most useful single sentence in Deloitte's 2026 Travel Industry Outlook is that generative AI use is rising in travel shopping "even as commerce and content are not fully connected," with **43% of high-income millennials who planned to travel using generative AI in trip planning** (a segment statistic, not market-wide) ([Deloitte Insights](#)). A conversational advisor that closes the plan-to-book loop addresses the gap Deloitte names.

What the firms actually run internally is instructive precisely because it is not voice. Deloitte is rolling Claude out to more than 470,000 employees with a Claude Center of Excellence and role-specialised variants ([CN BC](#)); PwC's agent OS is reported at 25,000 agents built with Salesforce, CrewAI and AWS; EY has 150 agents for 80,000 tax professionals with a target of 100,000 agents by 2028; KPMG Workbench mirrors audit-team oversight structures ([Enterprise DNA](#); [ChatFin](#), both third-party and unverified against firm press releases). The common architecture is a firm-owned orchestration layer above multiple licensed frontier models with human review preserved at the engagement level — de facto multi-vendor even though their research never says so. **None of the publicised internal deployments is a customer-facing voice system**; the architecture the Big Four have validated is text, internal, human-reviewed and latency-tolerant. Voice, external, real-time and unsupervised is a materially different risk profile that their own practice does not demonstrate.

Article 50, GDPR and Meta's rules already bind a WhatsApp voice agent

The binding AI Act hooks for a consumer-facing voice agent are narrower than the high-risk debate suggests, and they are already live. **Article 50(1) requires that systems interacting directly with people — explicitly "chatbots, AI agents, and avatars" — inform users they are dealing with AI, "clear, distinguishable and accessible and provided no later than the first interaction,"** and **Article 50(2) requires synthetic audio to be "marked in a machine-readable format and detectable as artificially generated"**; both apply from 2 August 2026, and the Digital Omnibus on AI (reported in force 27 July 2026, [UNVERIFIED]) pushed the high-risk deadlines to 2 December 2027 and 2 August 2028 but left

Article 50 untouched ([EC FAQ](#); [Goodwin](#); [Usercentrics](#)). One source reports a four-month transition for the machine-readable marking duty, to **2 December 2026, for systems already on the market as at 2 August 2026** — **[UNVERIFIED, secondary only, and contradicted in emphasis by sources that call Article 50 wholly untouched]** ([Usercentrics](#)). Penalties for Article 50 breaches are reported at up to €15m or 3% of worldwide turnover ([Greenberg Traurig](#)). A travel advisor is almost certainly not Annex III high-risk, so the deferral gives the build no breathing room — the Article 50 clock has already run out. Article 5's emotion-recognition prohibition (in force since 2 February 2025) is scoped to workplaces and education and to inference from biometric data such as voice tone, so it does not bite a consumer travel agent directly, but any emotion inference from voice still triggers Article 50 notification and is a trap worth avoiding ([FPF](#)). No harmonised standard identifies which audio watermark satisfies Article 50(2); standards were still delayed as of August 2026 ([Plesner](#)).

Under GDPR the design lever is the ASR/voiceprint split. The EDPB's position is that **"voice data is inherently biometric personal data,"** but Article 9 special-category status attaches only where processing is for uniquely identifying a person — speaker verification, not plain speech-to-text — and where it does attach, explicit consent from registered users plus an equivalent-functionality non-voiceprint alternative is required for consent to be freely given ([EDPB Guidelines 02/2021](#)). Those guidelines date from 2021 and predate LLM voice agents entirely; no updated EDPB guidance on generative voice agents was found. Spain's AEPD adds that biometric voice analysis can extract over a hundred parameters from which health conditions can be inferred ([AEPD](#)), and its **"Orientaciones sobre IA agéntica"** (**published 1 February 2026**) reads almost as an engineering checklist: **memory compartmentalisation, data-lifecycle traceability, meaningful human supervision, isolation environments, controlled updates, prompt-development SOPs, active DPO participation, and a DPIA "especially if automated decisions exist under Article 22"** ([AEPD](#); [Ecija](#)). Article 22 reaches further than it looks after **CJEU C-634/21 (SCHUFA, 7 December 2023)**: an automated output that plays a determining role in a downstream decision is itself a decision, and a human who rubber-stamps without genuine authority to reach a different conclusion does not break the chain ([TLT](#); [IAPP](#)). An agent whose answer in practice determines a fare, refund or EU261 outcome is exposed. The only concrete regulator retention figure found is CNIL's **six-month active retention for call recordings kept for quality control**, one year for derived analysis and five years for evidentiary recordings — a workplace call-recording context, so its transfer to a consumer AI channel is an extrapolation ([Axialys](#); [CNIL NS-57 PDF](#)). CNIL treats "providing a conversational assistance service" as a priori legitimate interest in the AI-development context ([CNIL](#)); the claim that CNIL requires a voice agent to identify itself "in the first seconds of a call" comes from a vendor blog and is **[UNVERIFIED]** ([Talkr.ai](#)).

Data residency constrains vendor choice in ways that matter for the reader's six-brain switch. **Anthropic's direct API is reported to offer only "us" and "global" inference geographies with no EU-only option, so guaranteed EU residency for Claude means routing through AWS Bedrock EU or Google Vertex AI EU regions** — which changes the DPA counterparty and audit story ([Eden AI](#), secondary; the "Covered Models" 30-day retention claim in the same source is **[UNVERIFIED]**). OpenAI offers in-region EU Projects with zero data retention ([Eden AI](#)). **ElevenLabs offers EU endpoints and a per-request Zero Retention Mode (`enable_logging=false`) that "doesn't cover voice cloning"** — the one operation most likely to involve Article 9 data is the one not covered ([Deepgram](#), competitor-authored). **No authoritative statement of Deepgram's EU residency or zero-retention terms was found at all**, and nothing on whether Meta offers EU residency for message content flowing through the Cloud API, even though Meta sits in the data path before any vendor. The EU–US Data Privacy Framework remains valid, but the General Court's dismissal of *Latombe* (3 September 2025) is under appeal as C-703/25 P, noyb is preparing "Schrems III," and the US Supreme Court's *Trump v. Slaughter* (29 June 2026) is reported to undercut the FTC-

independence premise the adequacy decision relies on ([EuropeanMartech](#); [activeMind.legal](#), [UNVERIFIED] on the ruling's details). The defensible posture is belt-and-braces: EU residency plus SCCs and a transfer impact assessment in addition to DPF certification, so a future invalidation is paperwork rather than architecture.

Meta's platform rules are the least stable layer. **In October 2025 Meta barred businesses whose primary offering is a general-purpose AI assistant from the WhatsApp Business Platform, effective 15 January 2026; the Commission opened an antitrust investigation, notified Meta of possible interim measures on 9 February 2026, and on 5 March 2026 Meta partially conceded a 12-month window for general-purpose assistants in Europe with a per-message fee** ([TechCrunch](#); [EC IP/26/310](#); [ALM Corp](#)). Structured bots for support, bookings and order tracking remain allowed ([respond.io](#)), so a travel advisor is on the right side of the line — but the line is Meta's to draw, and a conversational LLM lives in exactly the ambiguity between "structured booking bot" and "assistant-style open-ended interaction." Separately, **since January 2026 businesses must state that the user is interacting with an AI chatbot within the first message** ([Alibaba Cloud](#), secondary), which aligns with Article 50(1) so that one disclosure string can satisfy both — though Meta's is enforced by account suspension and the AI Act's by a regulator. The mechanics of audio are documented only through intermediaries: **16 MB size cap; a message renders as a true voice note only as OGG/Opus mono with MIME audio/ogg; codecs=opus**, otherwise it arrives as a file attachment ([Expertflow](#); [chatarchitect](#)); a single source claims the play icon appears only below **512 KB [UNVERIFIED]**, which if real caps a natural voice note at a minute or two and pushes the design toward short spoken replies plus text. Whether WhatsApp's transcoding preserves container metadata — and therefore whether a metadata-based Article 50(2) mark survives delivery — is unaddressed; an in-signal watermark is the more robust bet, to be tested empirically. The European Accessibility Act (transposed by 28 June 2025) explicitly reaches chatbots and helplines and requires that **where voice input is needed, at least one mode of operation must not require it** ([National Law Review](#); [Famulor](#)), which WhatsApp satisfies only if the text path is functionally equivalent — so no voice note should ever be the sole carrier of information. Finally, the product decision between *advising* and *selling* is also a licensing decision: organising or selling a combination of two or more travel services over 24 hours engages Atout France registration with financial guarantee and professional liability insurance under Code du tourisme L.211-18 in France, and RDL 23/2018 plus autonomic-community agency authorisation in Spain ([Atout France](#); [BOE-A-2018-17769](#)). IATA's own position that "extraordinary circumstances, multi-leg journeys and rebookings make blanket automation unworkable" for EU261 is a direct warning against confident entitlement answers ([IATA](#)).

Keep, change, add, avoid: applying the evidence to the current stack

The table below condenses the recommendations; the prose that follows explains the reasoning where it is not obvious.

Area	Keep	Change	Add	Avoid
Architecture	Async voice-note cascade; text at every boundary	Nothing structural now; treat streaming as a separate project gated on Calling API media access	Two model tiers (cheap live path, strong offline judge)	Native S2S as the primary path; building a WebRTC/SIP bridge before confirming the Calling API streams media

Area	Keep	Change	Add	Avoid
STT	Nova-3 and Scribe v2 as switchable; Whisper for async only	Benchmark on own accented ES/FR audio with airline jargon and PNRs, not vendor WER	A single-pass code-switching option (Universal-3.5 Pro or Speechmatics); Deepgram keyterm prompting for IATA/PNR; entity error rate as a first-class metric	Whisper on any future streaming path; trusting sub-2 s first turns for language routing
Brain	Six-way switch; IONOS Llama/Mistral as the genuinely EU-resident path	If EU residency is required for Claude, route via Bedrock EU or Vertex EU rather than the direct API	Per-locale formality pinned in the system prompt; template slots for every number, price, time and PNR	Free-form generation of fares, fees or EU261 entitlements
TTS	Aura-2 (7 languages incl. ES/FR) and ElevenLabs multilingual v2 for async	For any streaming path swap to a low-latency tier (Flash v2.5, Sonic-3.x)	OGG/Opus mono output; in-signal watermark; locale-aware number/date normalization; a text twin of every voice note	Voice cloning (outside zero retention); long spoken monologues
Translation	Text-level translation step	Placeholder-protect codes before MT, re-substitute after; lemma-level not verbatim glossaries for inflected terms	COMET plus LLM-judge on sampled turns; pivot ES ↔ FR via English if ever needed	Single-call speech-translate APIs that hide the intermediate text
Cost	Daily spend cap	Extend to per-conversation caps and per-language cost dashboards	Cost-per-resolved-conversation as the KPI	Planning around advertised \$0.05/min rates
Governance	—	Human escalation with authority to overrule, logged	First-turn AI disclosure as text and audio; consent via text buttons; tiered retention; per-turn audit trail; DPIA before processing	Emotion inference from voice tone; speaker verification/voiceprints
QA	—	—	Simulate-then-judge pre-release; 1–5% daily human sampling; per-language dashboard; PNR read-back in locale spelling alphabets	Aggregate-only metrics

Keep the cascade and the asynchronous mode. The reader's WhatsApp voice-note workload is half-duplex and asynchronous, so barge-in, sub-300 ms turn latency and semantic endpointing — the entire class of engineering that dominates 2026 voice-agent discourse — largely do not apply, and the researchers' explicit inference is that the reader can deprioritise it and reinvest in grounding and

transcription accuracy. The cascade also preserves reasoning under Big Bench Audio, gives structured tool contracts that gpt-realtime-2.1 cannot, and is cheaper at volume. The three-STT and three-TTS switches are the right posture in a commoditised component market, and the six-brain switch is de facto the multi-vendor strategy the Big Four practise but do not preach. The IONOS-hosted Llama 3.3 and Mistral path is the one option in the stack whose EU residency does not depend on a second-hand claim about a US vendor's geography.

Change the brain economics and the code path. The notes' recommended live-path defaults are small, fast models (GPT-4.1-mini; Claude Sonnet 4.5 where long context matters), with a stronger model held back for offline simulation and judging in the Parloa pattern; since LLM time-to-first-token dominates every latency budget and cost visibility is KPMG's strongest predictor of ROI, the reader should measure whether the default frontier brain earns its cost per resolved conversation on voice-note traffic, and route routine turns to a cheaper tier. Every figure the agent speaks — price, fee, time, PNR — should be a template slot filled from a GDS tool response, never LLM prose, and guardrails belong in the orchestration layer, not the prompt ([Softcery](#)). The alphanumeric problem is the most concrete change: implement extract-freeze-verify around PNRs and IATA codes, add read-back confirmation in the locale's spelling alphabet, and measure entity error rate separately from WER, because 85% mixed-alphanumeric accuracy means roughly one in seven record locators is wrong on first pass.

Add the governance items that are binding today, not scheduled. A single first-turn string — sent as text, repeated in the first seconds of any audio — covering that this is an AI agent, that the conversation may be transcribed and why, how to reach a human, and a privacy link, satisfies Article 50(1), Meta's first-message rule and the EAA screen-reader path at once; re-assert it whenever a new 24-hour window opens. Explicit consent for anything Article 9-adjacent should be captured through WhatsApp interactive buttons logged with timestamp and message ID, never by spoken "yes." Retention should run on separate clocks for raw audio (CNIL's six months as a ceiling, not a target), transcripts, derived evaluation sets and anything kept as booking evidence, with vector stores included because deleting from an index is a different operation from deleting a transcript. The audit trail should record per turn the model and version, system-prompt version, retrieval sources, tool calls, disclosure-sent and consent state, and any human override — which is what the AEPD's "data-lifecycle traceability" asks for and what takes the process out of Article 22(1) under *SCHUFA*. Automatic escalation triggers should include EU261 entitlement questions, refunds, accessibility requests, repeated failure and detected distress, and the human must have authority to change the outcome. A DPIA is triggered on at least three independent grounds and should precede processing.

Do not build the media bridge yet, and plan the numbers honestly if you do. The WhatsApp Business Calling signalling already in place is worth keeping, but the WebRTC/SIP media bridge should wait on three facts the research could not establish: whether the Calling API delivers a streaming media path to a machine agent, whether the reader's account clears the reported 2,000-conversation eligibility bar, and what the media path's codec, jitter and one-way delay look like. If those clear, the streaming build is a strictly smaller change than moving to S2S — keep the cascade's text boundaries, add Pipecat or LiveKit, Smart Turn v3 (23 languages, open source), a conversational STT with integrated endpointing (Flux Multilingual or Universal-3.5 Pro Realtime), and a Flash/Sonic-class TTS — with an **800 ms–1.2 s P95 target and a \$0.25–0.35/min all-in budget**, not the sub-500 ms and \$0.05/min of vendor marketing. On KPIs, adopt Hopper's "resolution over containment" framing and plan around **15–34% resolution** as the credible range for travel, treating ElevenLabs' 80% as a ceiling claim.

Avoid the traps that are specific to voice. Emotion inference from voice tone is biometric-based and, while outside the Article 5 workplace prohibition for a consumer product, adds an Article 50 disclosure duty

for negligible benefit. Speaker verification pulls the whole product into Article 9 explicit-consent territory with a mandatory non-biometric alternative. Voice cloning is excluded from ElevenLabs' zero-retention mode. Verbatim glossary constraints cause the very gender and number agreement errors they are meant to prevent in Spanish and French. And any drift toward "general assistant" behaviour on WhatsApp risks Meta's policy line, which has moved three times in twelve months and is scheduled to move again around March 2027.

What is still unknown

The gaps below are the ones that would most change the recommendations if resolved, and none of them could be closed with the sources reachable in this environment. First, **whether the WhatsApp Business Calling API exposes streaming audio to a machine agent** — no source addressed it, and it decides whether the streaming project is even possible on this channel. Second, **any vendor's actual Spanish or French word error rate**: AssemblyAI's FLEURS runs, the Open ASR Leaderboard's multilingual track and its paper all hold per-language tables that were blocked, and no independent head-to-head exists on peninsular versus Latin American Spanish or France versus Québec French, nor any IATA/PNR-specific recognition benchmark in either language. Third, **the liability of an automated advisor giving fare or ticketing guidance** — nothing was found on whether an AI fare quote binds the operator, on misrepresentation exposure, or on whether an AI agent can operate under an IATA-accredited agent's code; the researchers call this the weakest section of the entire brief. Fourth, the load-bearing legal dates that remain [UNVERIFIED]: the reported 2 December 2026 transition for Article 50(2) marking, the Digital Omnibus adoption sequence, Anthropic's "Covered Models" retention terms, and Deepgram's EU residency posture, which is entirely undocumented. Fifth, what Lufthansa actually deployed with Cognigy and whether any airline runs a production voice agent — the researchers report absence of evidence, not evidence of absence. Sixth, several benchmark conflicts that cannot be adjudicated from summaries: Flux's WER (6.99% vs 15.58% claims), Gemini 3.8 Live's hourly price (\$0.84 vs \$1.38), and gpt-realtime's real per-minute cost (\$0.06–0.11 cached vs \$0.25–0.35 all-in). Finally, two things nobody has studied: latency tolerance in asynchronous voice-note interaction, and how to detect the agent answering in the wrong language, for which the only proposed method — running language identification over the agent's own output — is the researchers' inference, not a sourced practice.

Conclusion

The most useful reframing this research offers is that the reader has, perhaps by accident, built the architecture the market is converging toward from the other direction. Real-time voice-agent vendors spent 2026 racing to fold turn detection into ASR, to add async tool calling to S2S models, and to recover the text boundaries that cascades have by construction — all to approximate the auditability and tool reliability the reader already has, while chasing a 600 ms floor that a controlled study suggests users barely reward beyond 1.5 s. The asynchronous voice-note channel is not a lesser version of a phone agent; it is a product with a different failure surface, where the binding constraints are alphanumeric fidelity, locale normalization, register, and a regulatory disclosure that must be delivered in text and audio simultaneously. The decisive question for the next twelve months is not "cascade or speech-to-speech" but whether Meta's Calling API ever gives a machine a media stream — and until it does, the highest-return engineering is a PNR read-back loop and a per-language dashboard, not a WebRTC bridge.

The second shift is in what "governance" means for a voice product. The Big Four's contribution turns out to be a cost argument dressed as a maturity argument: KPMG's 15%-versus-3% ROI split on cost visibility, and its finding that only a third of organisations can see what agents cost to run, point directly at the reader's spend cap as the most consequential feature in the stack — worth extending from a daily ceiling into a per-conversation, per-language cost-per-resolution instrument. Combined with the AEPD's agentic checklist and *SCHUFA*'s demand for a human who can actually change the answer, the regulatory and financial evidence converge on the same artefact: a per-turn audit trail that records what the agent cost, what it looked up, what it said, and who could have overruled it. Build that, verify the dates this report could not, and the rest of the stack can stay as it is.